Enterprise Journey To The Cloud

Beware of the speed breakers

BMK Lakshminarayanan

@LBMKRISHNA

ABOUT ME

BMK Lakshminarayanan

Solutions Architect | DevOps Advocate

MISSION

Make your life better,

Make others life better,

Make the world better

Community

DEVOPS DAYS

CLOUD NATIVE SUMMIT

CLOUD NATIVE COMPUTING FOUNDATION AMBASSADOR

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Cloud

DevOps

Microservices

Old ways
Won’t open
New doors

Governance

Cloud-Native

Canary Releases

Kubernetes

Pipeline

NoSQL

Statefulsets

Continuous Delivery

Event-Sourcing

Cloud

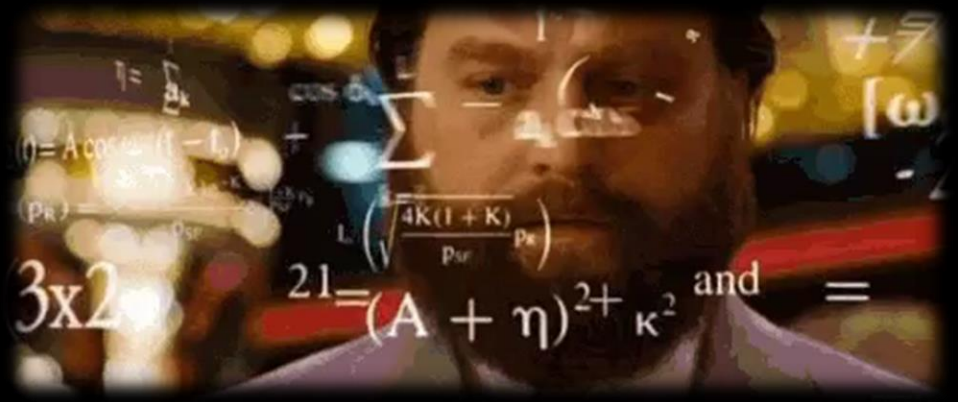
Hybrid-Cloud

Multi-Cloud

Containers

Canary Releases

yaml



Continuous Integration

Microservices

Lean

Observability

TDD

Open Source

DevOps

Scale

DevSecOps

Monoliths

DSC

A/B Testing

CQRS

Continuous Deployments

Agile

Operators

Dark Launch

API

Feature Toggles

PaaS

VSM

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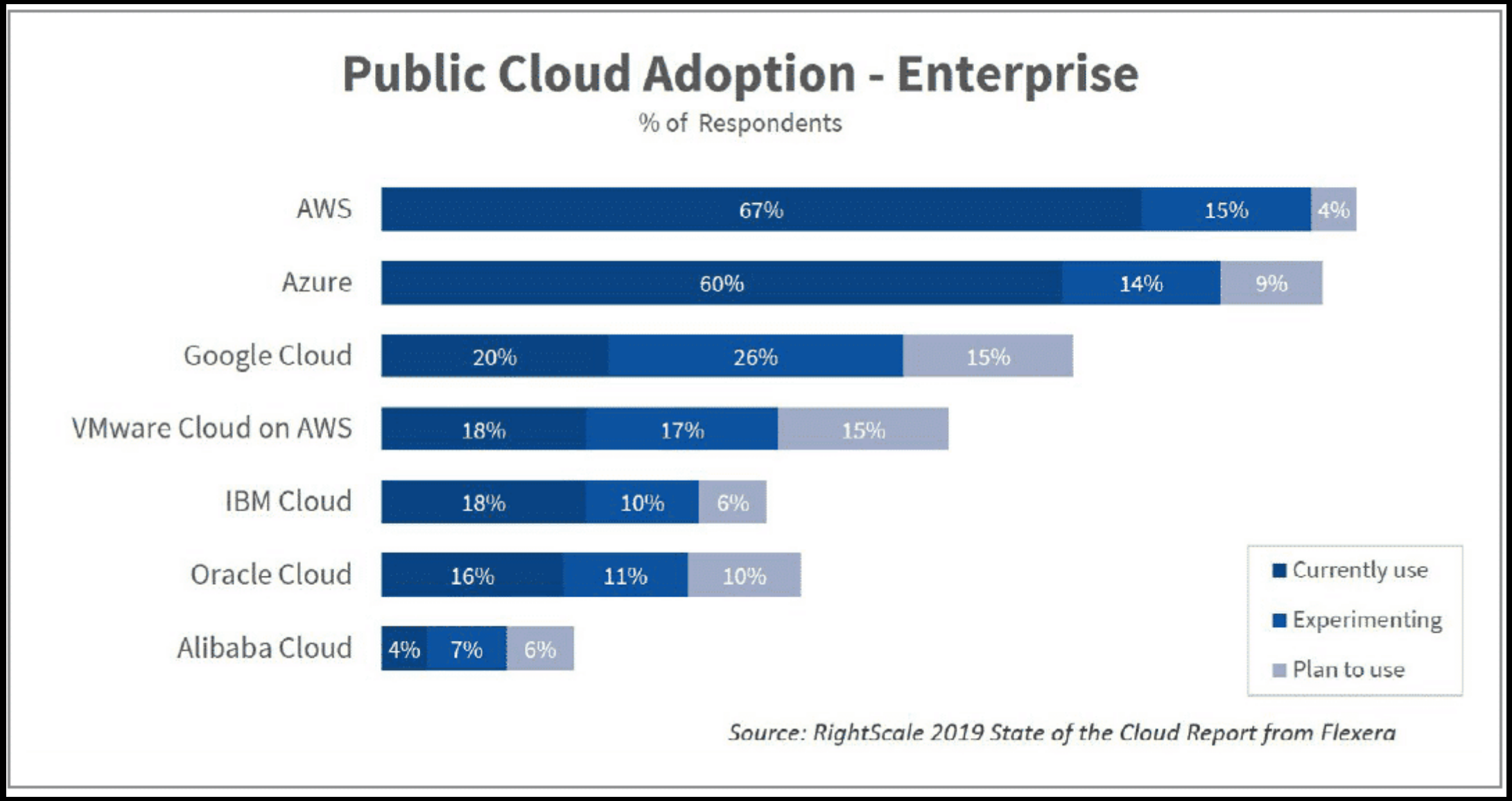
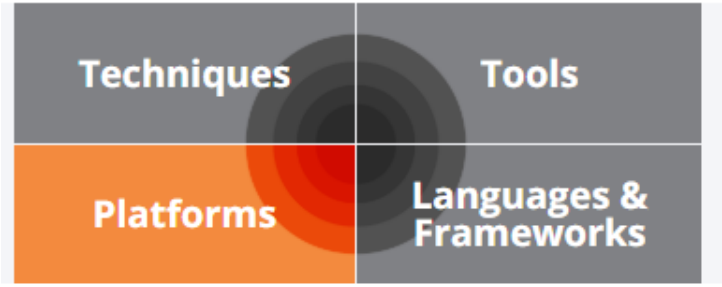
Enterprise Agility

Rapid and accelerating developments in technology, customer preferences, and employee expectations are transforming all aspects of our lives. To survive and thrive, companies must, in turn, transform how they operate.

https://www.mckinsey.com/


Public Cloud Adoption - Enterprise
% of Respondents
AWS
67%
15%
4%
Azure
60%
14%
9%
Google Cloud
20%
26%
15%
VMware Cloud on AWS
18%
17%
15%
IBM Cloud
18%
10%
6%
Oracle Cloud
16%
11%
10%
Alibaba Cloud
4%
7%
6%
Currently use
Experimenting
Plan to use
Source: RightScale 2019 State of the Cloud Report from Flexera



Platforms

Cloud

We feel strongly that the industry should be adopting these items. We use them when appropriate on our projects.

APR 2010 **ADOPT**

The Cloud continues to be of interest to us, with Software as a Service the most mature cloud component. Platform and Infrastructure as service offerings have reached different levels of maturity, and we reflect that in our placement of EC2, Google App Engine and Azure.

JAN 2010 **ADOPT**

ThoughtWorks®

https://www.thoughtworks.com/radar/platforms

Technology Radar

Cloud Challenges

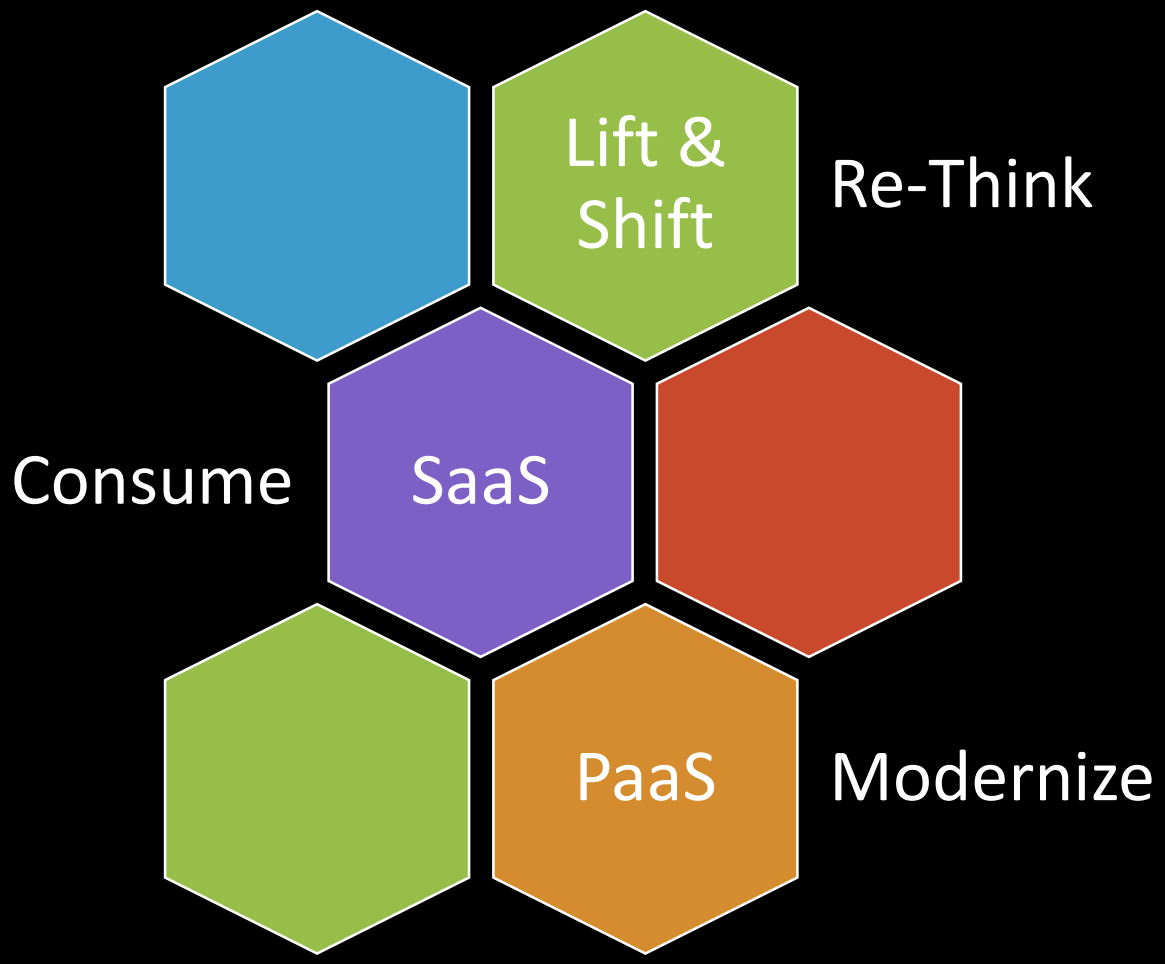
Cloud Challenges

Modernize or Leave behind or Lift & Shift?

Cloud-Native Architecture

Multi-Cloud

Lift & Shift, Leave Behind, Modernize: Apply Context, Understand “Why”

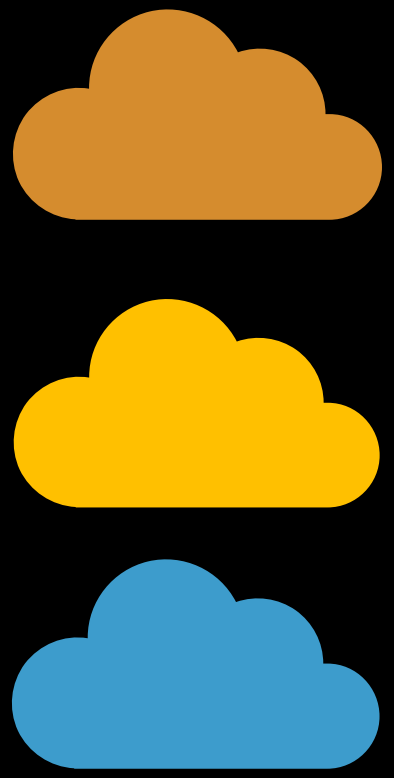


Cloud-Native Architecture?
Apply Modern Architecture Principles

https://www.cncf.io/

CNCF defines cloud-native as "scalable applications" running in "modern dynamic environments" that use technologies such as containers, microservices, and declarative APIs.

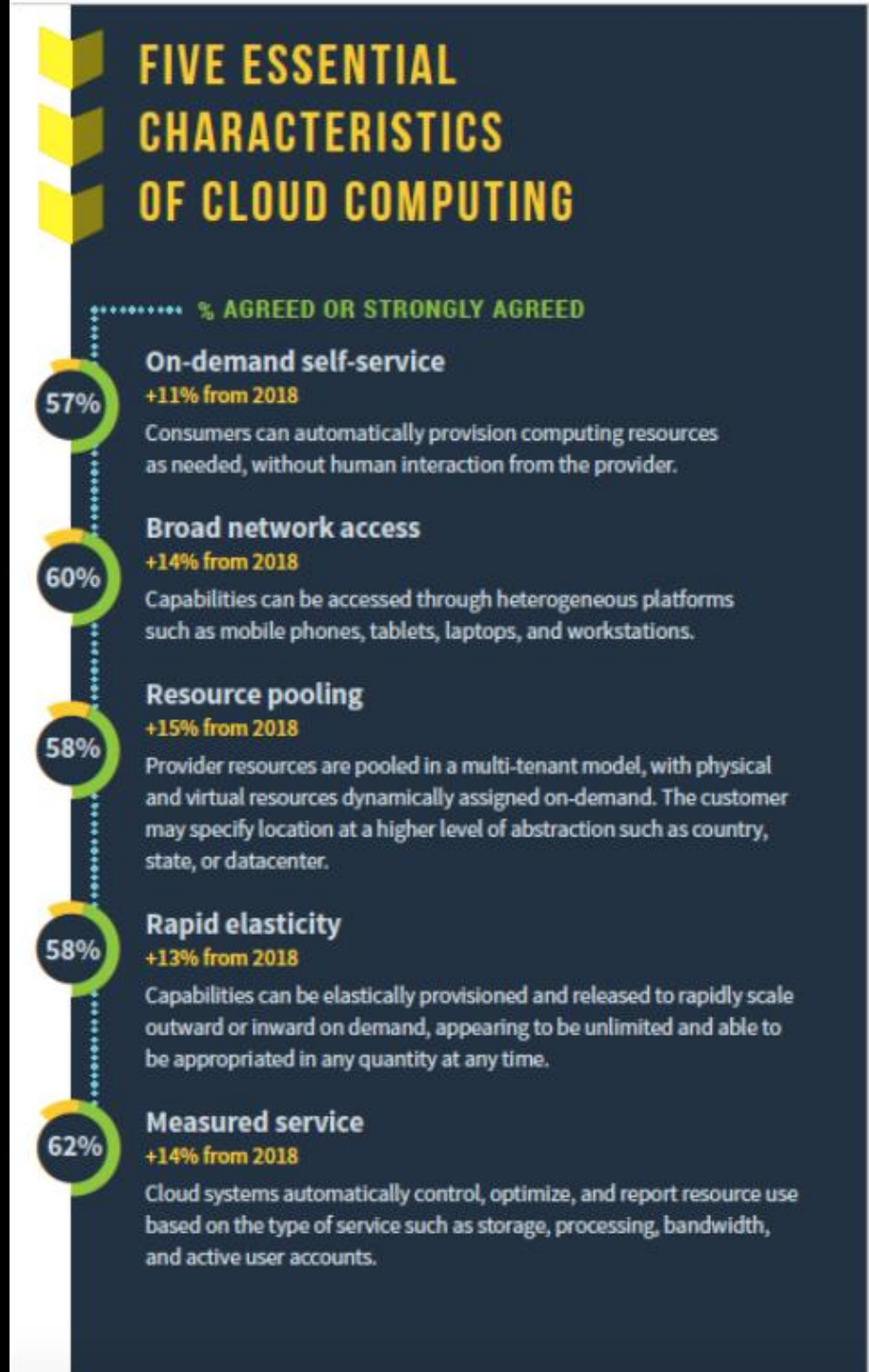
Multi-Cloud
Understand Business Risk, Opportunity & Continuity



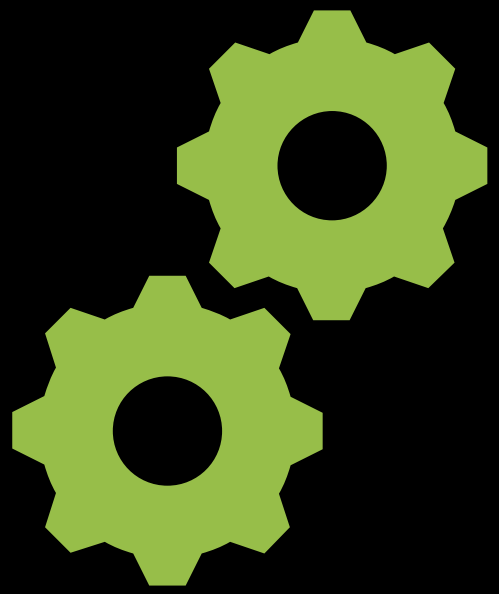
- Reducing reliance on any single vendor
- Cost-efficiencies
- Increasing flexibility through choice
- Adherence to local policies
- Mitigating against disasters.

Traps To Avoid

- Centralized “Cloud” Team
- Tickets to create “Cloud” resources
- “Cloud Governance”
- Security Restrictions



DEVOPS IS NOT A GOAL, BUT A NEVER-ENDING PROCESS OF CONTINUAL IMPROVEMENT
- Jez Humble



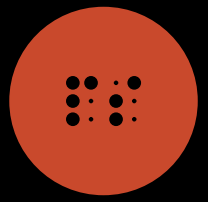
Culture
Automation
Lean
Metrics
Share

Some More Ideas

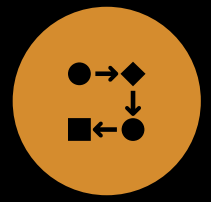
Ubiquitous

Disciplined & Focused Team

Sensible Architecture

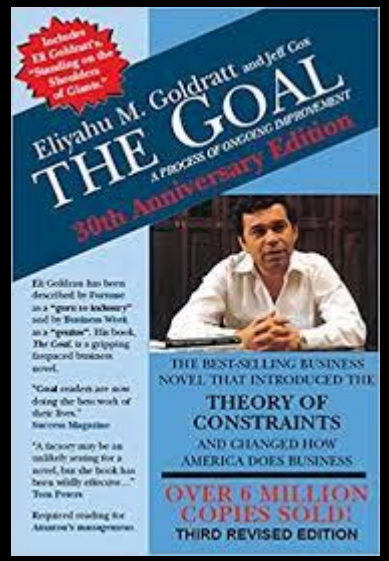


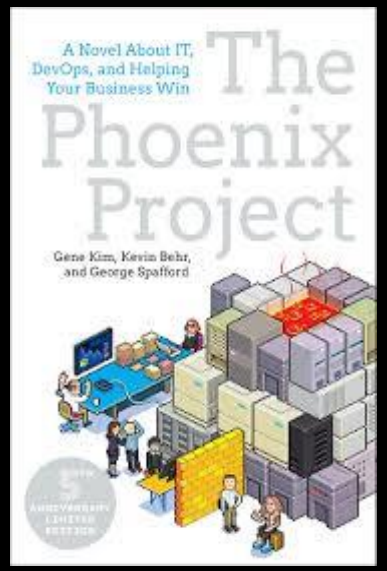
Upskill

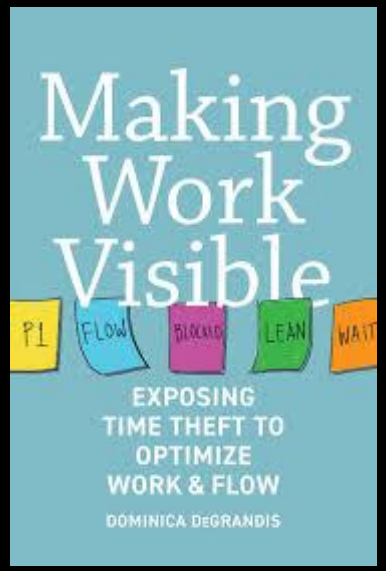


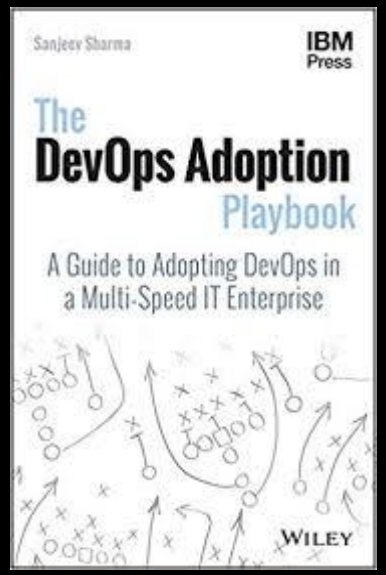
Reduce Context Switch

Developer Productivity

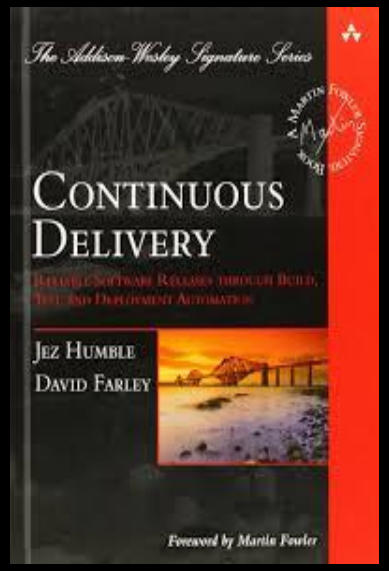

Eliyahu M. Goldratt and Jeff Cox
THE GOAL
A PROCESS OF ONGOING IMPROVEMENT
30th Anniversary Edition
THE BEST-SELLING BUSINESS NOVEL THAT INTRODUCED THE
THEORY OF CONSTRAINTS
AND CHANGED HOW AMERICA DOES BUSINESS
OVER 6 MILLION COPIES SOLD!
THIRD REVISED EDITION

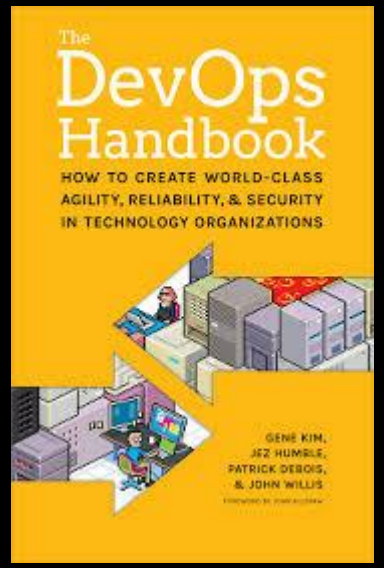

A Novel About IT, DevOps, and Helping Your Business Win
The Phoenix Project
Gene Kim, Kevin Behr, and George Spafford

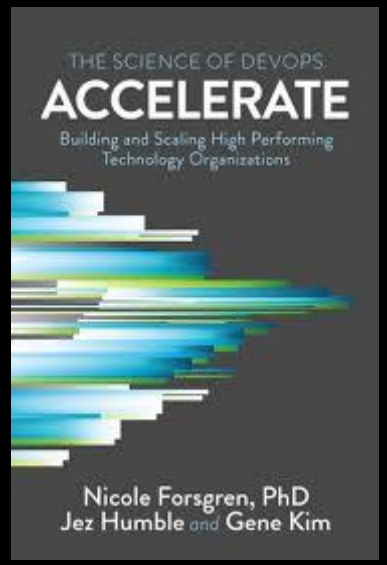

Making Work Visible
PI
FLOW
BLOCKED
LEAN
WAIT
EXPOSING TIME THEFT TO OPTIMIZE WORK & FLOW
DOMINICA DeGRANDIS

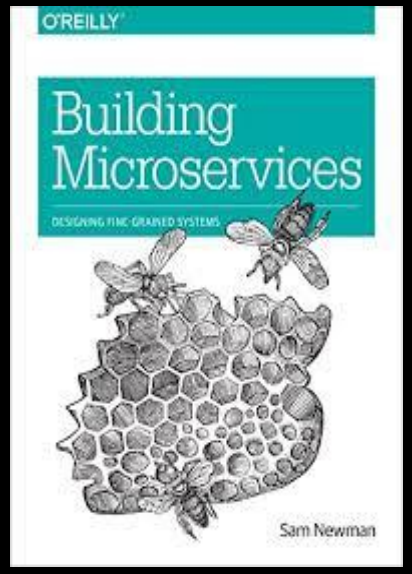

Sanjeev Sharma
IBM Press
The DevOps Adoption Playbook
A Guide to Adopting DevOps in a Multi-Speed IT Enterprise
WILEY

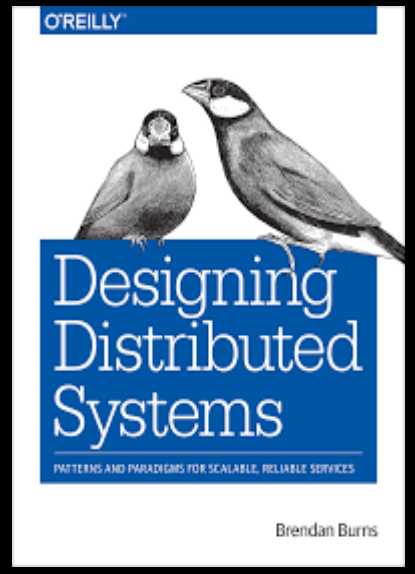
PROJECT TO PRODUCT
MIK KERSTEN
FOREWORD BY GENE KIM


The Addison-Wesley Signature Series
CONTINUOUS DELIVERY
JEZ HUMBLE
DAVID FARLEY
Foreword by Martin Fowler


The DevOps Handbook
HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, & SECURITY IN TECHNOLOGY ORGANIZATIONS
GENE KIM, JEZ HUMBLE, PATRICK DEBOIS, & JOHN WILLIS


THE SCIENCE OF DEVOPS
ACCELERATE
Building and Scaling High Performing Technology Organizations
Nicole Forsgren, PhD
Jez Humble and Gene Kim


O'REILLY
Building Microservices
DESIGNING FINE-GRAINED SYSTEMS
Sam Newman


O'REILLY
Designing Distributed Systems
PATTERNS AND PARADIGMS FOR SCALABLE, RELIABLE SERVICES
Brendan Burns

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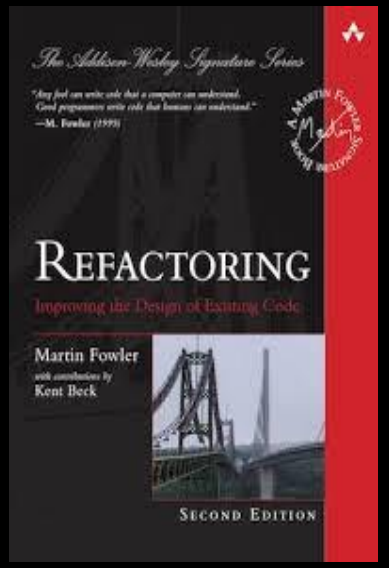

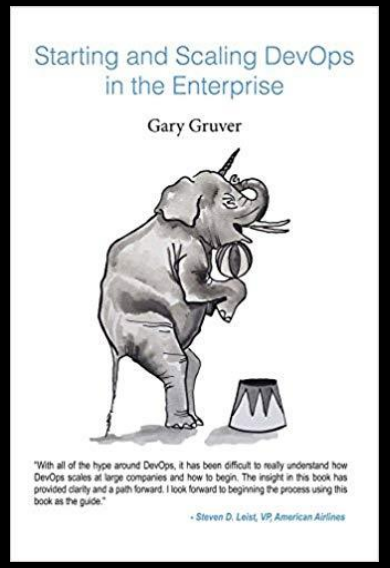

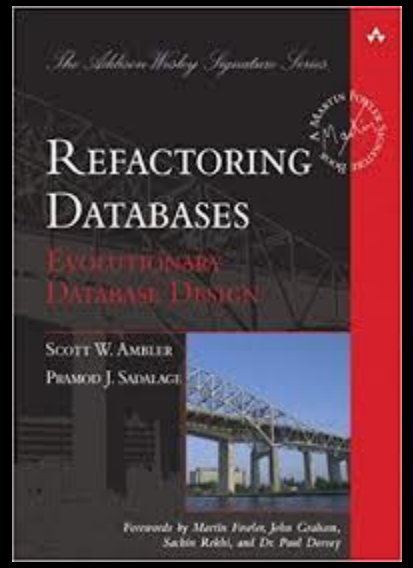

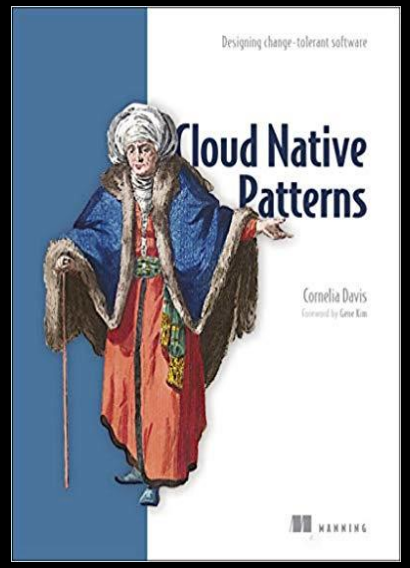

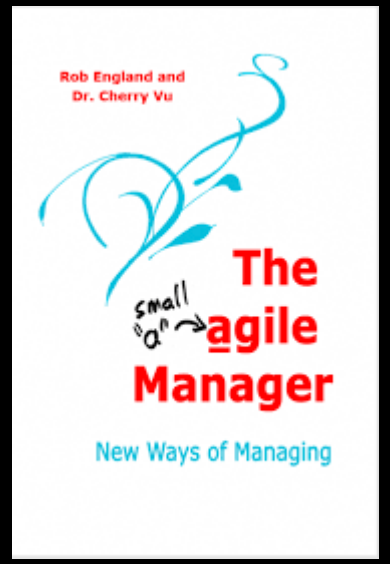

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DORA
DEVOPS RESEARCH & ASSESSMENT